The nature and variety of living organisms

10 questions: 3 multiple choice, 7 structured.

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1 · Part 1

Paper 1 & 2 Characteristics of living organisms [1]

Which list contains only characteristics that are shared by all living organisms?

A nutrition, photosynthesis, growth, excretion

B nutrition, respiration, movement, production of seeds

C respiration, cell walls, growth, response to surroundings

D excretion, reproduction, response to surroundings, nutrition

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2 · Part 1

Paper 1 & 2 Variety of living organisms [1]

Which row shows the features of the cells of a fungus such as *Mucor*?

	Chloroplasts	Cell wall	Carbohydrate store
A	present	made of cellulose	starch
B	absent	made of chitin	glycogen
C	absent	no cell wall	glycogen
D	absent	made of cellulose	starch

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3 · Part 1

Paper 1 & 2 Variety of living organisms [1]

Which statement about viruses is correct?

A Viruses can reproduce only inside living cells.

B Viruses have a cell membrane and cytoplasm.

C Viruses contain both DNA and RNA.

D Viruses are larger than bacteria.

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4 · Part 1

Paper 1 & 2 Characteristics of living organisms [6]

Robot vacuum cleaners clean floors without a person controlling them. The table describes four behaviours of one robot vacuum cleaner.

Behaviour of the robot	Characteristic of living organisms that it is most like
It travels around the floor of a room.	
When a sensor detects a wall, the robot changes direction.	
When its battery is low, it returns to its base and takes in electrical energy.	
When its motor gets too hot, a fan switches on and runs until the motor is back at its normal working temperature.	

(a) Complete the table by giving the characteristic of living organisms that each behaviour is most like. [4]

(b) Explain why the robot is not a living organism, even though it shows these behaviours. [2]

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5 · Part 1

Paper 1 & 2 | Variety of living organisms | [7]

Tissue samples were taken from a bean seed and from a piece of human liver. Each sample was tested with iodine solution. Iodine solution changes from orange-brown to blue-black when starch is present.

Sample	Colour after adding iodine solution
bean seed	blue-black
human liver	orange-brown

(a) State what these results show about each sample. [2]

(b) Bean plants and humans are both multicellular organisms. Give two differences, other than the carbohydrate they store, between the cells of a bean plant and the cells of a human. [2]

(c) Animals such as humans usually have nervous co-ordination. State one other feature that animals usually have but plants do not. [1]

(d) Sugar beet is a plant. Its roots store carbohydrate as sucrose. Suggest why testing a sugar beet root with iodine solution would not show that the root stores carbohydrate. [2]

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6 · Part 1

Paper 1 & 2 | Variety of living organisms | [7]

Mucor is a fungus. Its body is a network of thread-like structures, each containing many nuclei.

Ravi placed a small piece of *Mucor* in the centre of each of five dishes of agar jelly. The agar contained starch. He kept the dishes at 25 °C.

Each day he flooded one dish with iodine solution. Iodine solution turns blue-black when starch is present. The agar turned blue-black except in a clear circle around the *Mucor*. He measured the diameter of the clear circle.

Time / days	Diameter of clear circle / mm
0	0
1	6
2	14
3	22
4	30

(a) State the names of:

(i) the thread-like structures [1]

(ii) the network that these thread-like structures form. [1]

(b) Explain why a clear circle formed around the *Mucor*. [3]

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(c) Calculate the mean rate of increase in the diameter of the clear circle between day 1 and day 4.

Use the formula: rate = change in diameter ÷ time taken

Give your answer in mm per day. [2]

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7 · Part 1

Paper 1 & 2 | Variety of living organisms | [8]

Lactobacillus bulgaricus is a rod-shaped bacterium used to make yoghurt from milk. It produces lactic acid, which lowers the pH of the milk.

(a) *Lactobacillus bulgaricus* is a prokaryotic organism.

State one structure that a yeast cell has but a *Lactobacillus bulgaricus* cell does not have, and describe how the DNA in a *Lactobacillus bulgaricus* cell is arranged. [2]

(b) Priya added *Lactobacillus bulgaricus* to a flask of milk and kept it at 42 °C. She measured the pH of the milk every hour.

Time / hours	pH of milk
0	6.6
1	6.4
2	5.9
3	5.3
4	4.8
5	4.6
6	4.5

(i) Describe how the pH of the milk changed over the 6 hours. [2]

(ii) Calculate the mean rate of decrease in pH between 2 hours and 4 hours.

Use the formula: rate = change in pH ÷ time taken

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Give your answer in pH units per hour. [2]

(c) Priya also set up a second flask of milk with no bacteria added, kept at 42 °C. The pH of this milk stayed at 6.6.

Explain why she set up this second flask. [2]

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8 · Part 1

Paper 1 & 2 | Variety of living organisms [8]

(a) State what is meant by the term pathogen. [1]

(b) The table describes four pathogens, W, X, Y and Z, that cause disease in humans.

Pathogen	Description	Group
W	single-celled; has a nucleus; has no cell wall; lives inside human red blood cells	
X	single-celled; has no nucleus; has a circular chromosome of DNA and plasmids	
Y	made of a protein coat surrounding a strand of RNA	
Z	made of thread-like hyphae with cell walls made of chitin; grows on the skin between the toes	

Complete the table by deducing the group that each pathogen belongs to. Choose from: bacterium, fungus, protoctist, virus. [4]

(c) Tobacco mosaic virus infects tobacco plants. It prevents chloroplasts forming in the leaves.

Explain why tobacco plants infected with this virus grow less than healthy tobacco plants. [3]

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9 · Part 1

Paper 1 & 2 | Variety of living organisms | [7]

Influenza is caused by a virus. A scientist used an electron microscope to take an image of an influenza virus particle. In the image, the diameter of the virus particle is 24 mm. The magnification of the image is ×200 000.

(a) Calculate the actual diameter of the virus particle in µm.

Use the formula: actual size = image size ÷ magnification

1 mm = 1000 µm [2]

(b) *Pneumococcus* is a bacterium that causes pneumonia. One *Pneumococcus* cell has a diameter of 0.9 µm.

Calculate how many times greater the diameter of this *Pneumococcus* cell is than the diameter of the influenza virus particle. [2]

(c) Explain why viruses are not considered to be living organisms. [3]

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10 · Part 1

Paper 1 & 2 Variety of living organisms [7]

Euglena is a microscopic organism that lives in pond water. The list gives some of its features.

- It is single-celled.
- Its cell has a nucleus.
- Its cell contains chloroplasts and can carry out photosynthesis.
- Its cell has no cell wall.
- It swims through water using a whip-like structure.
- In the dark, it can take in small particles of food from the water.

(a) *Euglena* is classified as a protoctist.

Explain why *Euglena* is not classified as a bacterium, a plant, an animal or a fungus. Use the features in the list in your answer. [5]

(b) Suggest why being able to take in food particles in the dark helps *Euglena* to survive. [2]

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